



Introduction to Artificial Intelligence

Exploring the History, Concepts, and Theories Behind AI

Course: Artificial Intelligence

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Agenda:

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- Introduction to AI
- Defining Intelligence
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History of AI

- **1950s:** Birth of AI – Alan Turing and the Turing Test
- **1956:** Dartmouth Conference – The term "Artificial Intelligence" was coined
- **Artificial = Human-made**
- **1960s-70s:** Early AI research on problem-solving, game playing
- **1980s-90s:** Expert systems, machine learning begins
- **2000s-Present:** Deep learning, neural networks, AI in everyday applications (self-driving cars(Tesla), voice assistants(Siri , Alexa))

What is Intelligence?

- **Intelligence is the ability to learn, understand, and apply knowledge.**
- Ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience.
- **Artificial Intelligence** mimics/duplicate these cognitive/brain processes.

Characteristics of Intelligence

- **Learning:** Acquiring knowledge or skills from experience.
- **Reasoning:** Drawing conclusions based on information.
- **Problem-Solving:** Applying knowledge to solve unfamiliar/new situations.
- **Adaptability:** Changing actions based on new information or changing environments.
- **Creativity:** Producing unique solutions or ideas.

Perspectives in AI on Intelligence

AI pioneering theorists

- **Alan Turing:** Intelligence as the ability to exhibit behavior indistinguishable from a human. **(Behave-like human)**
- **John McCarthy:** Intelligence involves using reasoning to perform tasks. **(Reasoning & logical thinking)**
- **Marvin Minsky:** Intelligence is the ability to solve problems that were not initially programmed. **(Can solve new problem)**
- **Herbert Simon:** Focus on problem-solving and decision-making as the essence of intelligence. **(Problem-solving & decision-making)**

Characterizing AI Systems

- **Thinking vs. Acting:**
 - **Thinking:** Cognitive/brain processes like reasoning and decision-making.
 - **Acting:** Practical application of those processes, e.g., robotics, automation.
- **Examples:**
 - **Thinking:** Chess-playing algorithms, natural language processing.
 - **Acting:** Autonomous vehicles, robotic systems.

Cognitive Science and AI

- Cognitive Science is the study of mind and intelligence, bridging psychology, neuroscience, artificial intelligence, philosophy, and linguistics.
- AI systems are inspired by human cognition, but they are not identical.
- Cognitive models, like neural networks, aim to simulate human brain processes.

Rational vs. Human Intelligence

- **Rational Intelligence:** Based on logic and formal rules (e.g., problem-solving algorithms).
- **Human Intelligence:** Includes emotions, social understanding, creativity, and experience-based learning.

Common Misconceptions about AI

- **AI equals human-level intelligence:** AI is specialized and not general intelligence.
- **AI is infallible/Error-Free:** AI can make mistakes, especially when faced with incomplete or biased data.
- **AI will replace all jobs:** AI will complement/Enhance/Boost human work, not replace it entirely.

Assumptions/Belief Behind AI Research

- Algorithms can learn from data
- Machines can simulate/duplicate human cognitive processes.

Goals of AI Research

- **Engineering Goal:** Build systems that perform tasks at human or superhuman levels (e.g., self-driving cars, medical diagnostics).
- **Scientific Goal:** Understand human cognition, learning, and intelligence.

Is AI Possible?

- **Yes, to a degree:** Narrow AI (or Weak AI) already exists and excels in specialized tasks.
- **General AI (AGI):** A machine with the ability to understand or learn any intellectual task that a human can – still a long-term goal.
- **Challenges:** Human cognition involves emotions, social contexts, and experiences, making AGI more complex than narrow AI.

Turing Test

- Proposed by Alan Turing in 1950 as a test of a machine's ability to exhibit intelligent behavior indistinguishable from a human.
- A machine is said to "think" if a human evaluator cannot reliably distinguish between the machine and human responses in a conversation.
- **Limitations:** It tests mimicry of human behavior, not the true nature of intelligence.

Thanks